Exercise – Understanding maximum simulated likelihood theory

1. Aim of the exercise

The objective of this exercise is to build a unified theoretical understanding of simulation-based estimation methods by synthesizing the core frameworks of Train (2003) and Stern (2010). Specifically, we bridge Stern's generalized mathematical approach to expectation integrals with Train's classic formulation of discrete choice mixtures. By working through this connection, we will learn how abstract multi-dimensional integration challenges are translated into computationally tractable Maximum Simulated Likelihood (MSL) problems, establishing the foundations necessary to estimate complex, structural lifecycle dynamics using real-world panel data.

2. The mathematical problem

Before developing structural econometric models, we must establish the core mathematical problem that makes simulation-based estimation necessary. This section outlines the properties of expectation integrals that lack closed-form analytical solutions, evaluates why traditional numerical integration schemes fail under high dimensions, and introduces the standard Monte Carlo frequency simulator as a baseline engine to approximate these intractable expressions.

2.1. The integration problem

The foundational challenge in simulation-based inference is the evaluation of the expected value of a behavioral or statistical function. Following the generalized framework of Stern (2010), let U denote a vector of random variables with joint density $f\left(u \mid \theta\right)$, where θ is a vector of structural parameters. If $h\left(U\right)$ maps this random vector into an economic outcome, its expectation is given by

$$\mathbb{E}\left[h\left(U\right)\right] = \int h\left(u\right) f\left(u \mid \theta\right) \mathrm{d}u \tag{1}$$

In empirical econometrics, computing choice probabilities, moments, or individual expectations amounts to evaluating precisely this integral.

2.2. The curse of dimensionality

When the expectation admits a closed-form solution, no numerical approximation is required. When a closed-form expression is not available and the dimension of the random vector U is low (typically fewer than four dimensions), classical numerical integration methods, such as Gauss-Hermite quadrature or Newton-Cotes formulas, provide highly accurate and computationally efficient approximations. However, when modeling complex multi-period lifecycles or multinomial discrete choices with many alternatives, the integration space expands. Because the computational burden of traditional quadrature methods scales exponentially with the number of dimensions, these deterministic approximations rapidly break down. This computational barrier demands a shift toward simulation-based integration methods.

2.3. The Monte Carlo simulator

To bypass the curse of dimensionality, we replace the analytical integral with a simulated sample average. By drawing R i.i.d. realizations of the random vector from the density $f\left(u \mid \theta\right)$, denoted as $\{u^1, u^2, \ldots, u^R\}$, we construct the standard Monte Carlo integration engine:

$$\hat{\mathbb{E}}\left[h\left(U\right)\right] = \frac{1}{R}\sum_{r=1}^{R} h\left(u^r\right) \tag{2}$$

Because each simulated draw u^r is generated from the true density $f(u \mid \theta)$, we have

$$\mathbb{E}[h\left(u^r\right)] = \mathbb{E}\left[h\left(U\right)\right] \tag{3}$$

for every r. This equality ensures that the sample average reproduces the correct expectation operator, making the frequency simulator an unbiased estimator of the target integral for any fixed sample size, meaning

$$\mathbb{E}\left[\hat{\mathbb{E}}\left[h\left(U\right)\right]\right] = \mathbb{E}\left[h\left(U\right)\right]. \tag{4}$$

3. Maximum simulated likelihood (MSL) theory

Once we accept that certain structural expressions must be approximated via simulation, we must evaluate how these approximations affect our estimation routines. This section introduces the MSL framework, details the inherent log-transformation bias introduced by nested simulators via Jensen's Inequality, and outlines the strict asymptotic rules governing sample size and draw repetitions required to ensure consistent and efficient parameter estimates.

3.1. The MSL objective function

Suppose we observe a sample of N independent individuals indexed by $i = 1, \ldots, N$. Let the true probability of observing individual i's chosen behavioral outcome, given a vector of structural parameters θ, be denoted by $\mathrm{P}_i(\theta)$. Because individual choices depend on unobserved random utility shocks $U_i \sim f(u \mid \theta)$, this probability is equal to the expected value of a binary choice indicator function $h(U_i)$:

$$\mathrm{P}_i\left(\theta\right) = \mathbb{E}\left[h\left(U_i\right)\right] = \int h\left(u\right) f\left(u \mid \theta\right) \mathrm{d}u, \tag{5}$$

where $h(U_i) = 1$ if the unobserved shock vector U_i causes the observed choice to be utility-maximizing, and $h(U_i) = 0$ otherwise.

When the integral in (5) cannot be solved analytically and high dimensionality rules out standard quadrature, evaluating the true choice probabilities $\mathrm{P}_i(\theta)$ is impossible. To overcome this, we approximate $\mathrm{P}_i(\theta)$ by drawing R independent realizations of the unobserved shocks, $\{u_i^1, u_i^2, \ldots, u_i^R\}$, from the density $f(u \mid \theta)$ for each individual i. We then construct the frequency simulator as a sample average of these binary choices:

$$\hat{\mathrm{P}}_i\left(\theta\right) = \frac{1}{R}\sum_{r=1}^{R} h\left(u_i^r\right). \tag{6}$$

Because $\mathbb{E}\left[\hat{\mathrm{P}}_i\left(\theta\right)\right] = \mathrm{P}_i(\theta)$, $\hat{\mathrm{P}}_i\left(\theta\right)$ provides an unbiased simulator of the true choice probability.

With an engine in place to simulate individual choice probabilities, we turn to estimating the unknown parameter vector θ for the sample. In standard Maximum Likelihood Estimation, the objective is to maximize the sample log-likelihood function:

$$L(\theta) = \sum_{i=1}^{N} \ln \mathrm{P}_i(\theta). \tag{7}$$

However, because the exact choice probability $\mathrm{P}_i(\theta)$ lacks a closed-form solution, evaluating $L(\theta)$ directly is impossible. To overcome this computational barrier, we replace the intractable theoretical probability $\mathrm{P}_i(\theta)$ in equation (7) with its Monte Carlo simulator $\hat{\mathrm{P}}_i(\theta)$. This substitution yields the simulated log-likelihood function $l(\theta)$:

$$l(\theta) = \sum_{i=1}^{N} \ln \hat{\mathrm{P}}_i(\theta). \tag{8}$$

The MSL estimator $\hat{\theta}_{\text{MSL}}$ is the specific parameter vector that maximizes this simulated objective function:

$$\hat{\theta}_{\text{MSL}} = \arg\max_{\theta} l(\theta). \tag{9}$$

3.2. The log-transformation bias

A fundamental theoretical complication arises when moving from the probability simulator in equation (6) to the simulated log-likelihood function in equation (8). While $\hat{\mathrm{P}}_i(\theta)$ is an unbiased simulator of $\mathrm{P}_i(\theta)$, the natural logarithm is a strictly concave transformation. By Jensen's Inequality, taking the expected value of a concave function of a random variable yields a value strictly less than the function of its expected value:

$$\mathbb{E}\left[\ln \hat{\mathrm{P}}_i(\theta)\right] < \ln\left(\mathbb{E}\left[\hat{\mathrm{P}}_i(\theta)\right]\right). \tag{10}$$

Because the probability simulator is unbiased such that $\mathbb{E}\left[\hat{\mathrm{P}}_i(\theta)\right] = \mathrm{P}_i(\theta)$, substituting this equality into equation (10) yields:

$$\mathbb{E}\left[\ln \hat{\mathrm{P}}_i(\theta)\right] < \ln \mathrm{P}_i(\theta). \tag{11}$$

This inequality reveals that the simulator introduces an inherent, downward simulation bias into the log-likelihood function. For a fixed number of draws R, the objective function being maximized is fundamentally distorted, meaning that standard maximum likelihood properties do not automatically hold.

3.3. Asymptotic convergence rules

To achieve consistency and asymptotic efficiency, the simulation noise must be asymptotically negligible relative to the sampling variance as the sample size expands. Formally, the variance of the true maximum likelihood estimator is governed by the sampling variance of order $\mathcal{O}\left(\frac{1}{N}\right)$. In contrast, the Monte Carlo approximation introduces an additional simulation variance of order $\mathcal{O}\left(\frac{1}{NR}\right)$.

For the slope of the simulated log-likelihood function (the simulated score) to match the true slope, the extra simulation error — which scales as $O_p\left(\frac{\sqrt{N}}{R}\right)$ — must shrink to zero. This requires the number of Monte Carlo draws R to grow faster than $\sqrt{N}$ (i.e., $\frac{R}{\sqrt{N}} \to \infty$ as $N \to \infty$). Under this condition, the extra simulation noise disappears, making the MSL estimator perform identically to the true maximum likelihood estimator.

Under standard regularity conditions and the Central Limit Theorem, the asymptotic properties of the MSL estimator $\hat{\theta}_{\text{MSL}}$ are governed by the joint limiting behavior of the sample size N and the number of simulation draws R.

First, for consistency, the estimator $\hat{\theta}_{\text{MSL}}$ converges in probability to the true parameter vector θ_0 provided that both $N \to \infty$ and $R \to \infty$. So long as R grows without bound alongside N, the simulation bias vanishes asymptotically, ensuring $\operatorname{plim} \hat{\theta}_{\text{MSL}} = \theta_0$.

Second, while consistency only requires $R \to \infty$, achieving asymptotic efficiency and standard limiting normality demands a stricter condition. The MSL estimator achieves the exact same asymptotic efficiency as the infeasible Maximum Likelihood estimator if and only if R increases at a rate strictly faster than $\sqrt{N}$. Formally, this requires:

$$\frac{\sqrt{N}}{R} \to 0 \quad \text{as} \quad N, R \to \infty. \tag{12}$$

If R increases too slowly relative to N (failing to outpace $\sqrt{N}$), residual simulation bias introduces an asymptotic mean shift into the limiting normal distribution, distorting standard errors and invalidating conventional hypothesis testing.

4. An illustration: The Mixed Logit model

With the general asymptotic properties of the MSL estimator established, we illustrate this machinery using a prominent econometric specification: the Mixed Logit model. This section demonstrates how incorporating unobserved taste heterogeneity relaxes the restrictive IIA assumption, explicitly maps Train's continuous probability mixture framework onto Stern's generalized expectation formulation, and details the extension to multi-period panel choice sequences.

4.1. Bypassing Independence of Irrelevant Alternatives

Standard multinomial logit models operate under the assumption that unobserved disturbances are independent and identically distributed across alternatives. This error structure leads directly to the Independence of Irrelevant Alternatives (IIA) property. IIA means that when we take the ratio of choice probabilities for any two alternatives, i and k, the denominator containing all other alternatives cancels out completely:

$$\frac{\mathrm{P}_{ni}}{\mathrm{P}_{nk}} = \frac{\frac{\exp(X_{ni}\beta)}{\sum_j \exp(X_{nj}\beta)}}{\frac{\exp(X_{nk}\beta)}{\sum_j \exp(X_{nj}\beta)}} = \frac{\exp(X_{ni}\beta)}{\exp(X_{nk}\beta)} = \exp\left((X_{ni} - X_{nk})\beta\right) \tag{13}$$

Notice that the right-hand side depends only on the attributes of options i and k. The presence, removal, or quality of any third option j is completely missing. As a result, if a third option changes, the model can only keep this ratio $\frac{P_{ni}}{P_{nk}}$ constant by forcing the choice probabilities of all existing options to scale up or down by the exact same percentage, a rigid restriction known as proportional substitution that rarely holds in realistic economic settings.

The Mixed Logit model relaxes this limitation by allowing taste parameters β to vary randomly across individuals according to a continuous probability density $f(\beta \mid \theta)$. To find the overall (unconditional) choice probability for an individual drawn at random from the population, we integrate over all possible values of β:

$$\mathrm{P}_{ni}(\theta) = \int \left[\frac{\exp(X_{ni}\beta)}{\sum_j \exp(X_{nj}\beta)} \right] f\left(\beta \mid \theta\right) \mathrm{d}\beta. \tag{14}$$

Taking the ratio of unconditional probabilities between options i and k yields:

$$\frac{\mathrm{P}_{ni}(\theta)}{\mathrm{P}_{nk}(\theta)} = \frac{\int \left[\frac{\exp(X_{ni}\beta)}{\sum_j \exp(X_{nj}\beta)} \right] f\left(\beta \mid \theta\right) \mathrm{d}\beta}{\int \left[\frac{\exp(X_{nk}\beta)}{\sum_j \exp(X_{nj}\beta)} \right] f\left(\beta \mid \theta\right) \mathrm{d}\beta}. \tag{15}$$

Because the summation $\sum_j \exp(X_{nj}\beta)$ is trapped inside the integrals, it can no longer be canceled out. The ratio now explicitly depends on the attributes of every alternative j in the choice set. This integration captures unobserved taste heterogeneity across the population, freeing the model from fixed relative ratios and generating flexible, realistic substitution patterns.

4.2. Mapping to the general integrand

To illustrate how the Mixed Logit maps onto the general mathematical foundations established in Module 1, let the utility of individual n choosing alternative i be given by

$$U_{ni} = X_{ni}\beta_n + \epsilon_{ni} \tag{16}$$

where ϵ_{ni} is independent extreme value. The parameter vector β_n varies across individuals according to a continuous density function $f\left(\beta \mid \theta\right)$.

Conditional on knowing a specific realization of β_n, the choice probability collapses to the standard, closed-form logit fraction:

$$L_{ni}\left(\beta\right) = \frac{\exp\left(X_{ni}\beta\right)}{\sum_j \exp\left(X_{nj}\beta\right)} \tag{17}$$

Because β_n is unobserved, the unconditional choice probability must integrate out this variation across the entire taste distribution:

$$\mathrm{P}_{ni}\left(\theta\right) = \int L_{ni}\left(\beta\right) f\left(\beta \mid \theta\right) \mathrm{d}\beta \tag{18}$$

This formula maps exactly onto Stern's foundational expectation model,

$$\mathbb{E}\left[h(U)\right] = \int h\left(u\right) f\left(u \mid \theta\right) \mathrm{d}u \tag{19}$$

by setting the behavioral mapping function as $h(u) = L_{ni}(\beta)$ and the structural integration vector as $u = \beta$.

4.3. Panel data setup

In longitudinal settings where individual n is observed making a sequence of choices over multiple periods $t = \{1, 2, \ldots, T\}$, the framework adapts cleanly to accommodate persistent unobserved factors. Assuming that the individual's taste vector β_n remains stable over time, their choices across periods are conditionally independent given that specific draw.

Let $\mathbf{i} = (i_{n1}, i_{n2}, \ldots, i_{nT})$ represent the observed sequence of choices for individual n. The joint probability of observing this exact choice sequence is the product of the period-specific logit fractions integrated over the parameter density:

$$\mathrm{P}_{n\mathbf{i}}(\theta) = \int \left[\prod_{t=1}^{T} \frac{\exp(X_{ni_{nt}t}\beta)}{\sum_j \exp(X_{njt}\beta)} \right] f(\beta \mid \theta)\, \mathrm{d}\beta \tag{20}$$

Applying the Monte Carlo simulator from Section 2.3, we generate R draws from $f(\beta|\theta)$ to approximate this multi-period sequence probability:

$$\hat{\mathrm{P}}_{n\mathbf{i}}(\theta) = \frac{1}{R} \sum_{r=1}^{R} \left[\prod_{t=1}^{T} \frac{\exp(X_{ni_{nt}t}\beta^r)}{\sum_j \exp(X_{njt}\beta^r)} \right] \tag{21}$$

This simulated probability $\hat{\mathrm{P}}_{n\mathbf{i}}(\theta)$ is ready to be embedded directly into the MSL objective function. In empirical settings, the linear utility specification $X_{nit}\beta$ can be seamlessly replaced with structural preference functions-such as lifetime consumption and leisure allocations-to estimate complex lifecycle dynamics using real-world data.

5. Final notes

This exercise draws on three references. The first is: Train, K., 2003. Discrete Choice Methods with Simulation. Cambridge University Press. Available at: link. The second is: Stern, S., 2010. Simulation-based inference in econometrics: motivation and methods. In: Mariano, R.S., Schuermann, T., Weeks, M.J. (Eds.), Simulation-Based Inference in Econometrics: Methods and Applications. Cambridge University Press, pp. 9-38. Available at: link.